

Physical Properties of Rocks and How We Estimate Them

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1 Introduction

In geophysics we typically *measure fields* and *infer properties*, the reverse of many classical physics problems. Rock properties arise from minerals and the fluids in pore space and depend on the *state* of the system (pressure, temperature, volume, composition).

1.1 State variables

- Pressure P (Pa, MPa, GPa, kbar). Note: 1 kbar $\approx$ 100 MPa.
- Temperature T (K or $^{\circ}\text{C}$). Absolute scale: not “degrees K”; conversion: $T_K = T_{^{\circ}\text{C}} + 273.15$.
- Volume V (m^3 or km^3).
- Composition (moles, wt%, ppm, fractions). Can refer to phase modes or elemental/oxide concentrations.

2 Key Physical Properties

- Density: $\rho = m/V$ (kg m^{-3}).
- Thermal expansivity (volumetric, at P const.): α (K^{-1}).
- Compressibility β_T and bulk moduli $K_T = 1/\beta_T$, K_S (MPa, GPa).
- Shear modulus μ (MPa, GPa).
- Seismic velocities: v_p (compressional), v_s (shear) (m s^{-1} or km s^{-1}).
- Thermal conductivity k ($\text{W m}^{-1} \text{K}^{-1}$); heat capacity c_p ($\text{J kg}^{-1} \text{K}^{-1}$).
- Thermal diffusivity $\kappa = k/(\rho c_p)$ ($\text{mm}^2 \text{s}^{-1}$ or $\text{km}^2 \text{Ma}^{-1}$).
- Heat production A (W m^{-3} or $\mu\text{W m}^{-3}$), or H (W kg^{-1}).

3 Strategies to Estimate Properties

3.1 Direct measurements

- Lab or in situ; strong control on state; targeted protocols.
- Consider scaling, representativeness, cost, and sampling.

3.2 From composition

- From mineral modes: simple to apply when modes known; extrapolation outside calibrations is risky (??).
- From major elements: widely available; captures broad trends but may obscure mineral-control (assumptions needed).
- Thermodynamics: internally consistent; works with or without known modes; computationally intensive and specialized.

3.3 Indirect correlations

- Property–property regressions (e.g., v_p vs ρ); rapid domain mapping but potentially non-unique.
- Data-driven/ML: handles large, multivariate datasets; may be biased by training data and limited in state extrapolation.

4 Examples and Figures from Notes

4.1 Density from composition

An empirical relation for many silicates (Group I) is suggested as

$$\rho [\text{kgm}^{-3}] = 2532 + 216\text{Fe} + 608M - 10\text{MALI}, \quad (\text{notation as in notes; confirm definitions: ??}) \quad (1)$$

where Fe is an iron number (wt%), M a maficity parameter, and MALI is the modified alkali–lime index (precise definitions deferred: ??).

4.2 Other property trends

Several properties co-vary with composition or with each other (e.g., thermal conductivity with composition; weakly with v_p), enabling indirect estimates.

5 Pressure, Temperature, and Porosity Effects

For a given rock, three principal factors influence velocity:

- Pressure increases close cracks and stiffens the frame $\Rightarrow v$ increases.
- Temperature increases reduce density and moduli $\Rightarrow v$ decreases.
- Porosity/fractures impede wave paths $\Rightarrow$ lower apparent v (in crystalline rocks, density reduction may be minor).

6 Mixing Relations

Rocks are mixtures; bulk properties require mixing laws.

6.1 Common means

- Arithmetic mean (e.g., density, some heat capacities): $X = \sum_i w_i X_i$.
- Geometric mean (random mixtures; often used for conductivity): $X = \prod_i X_i^{w_i}$.
- Harmonic mean (series layers; transport-limited): $X = \left(\sum_i \frac{w_i}{X_i} \right)^{-1}$.

6.2 Voigt–Reuss–Hill (VRH)

For elastic moduli/velocities: X_V (arithmetic), X_R (harmonic), and Hill average $X_{\text{VRH}} = \frac{1}{2}(X_V + X_R)$.

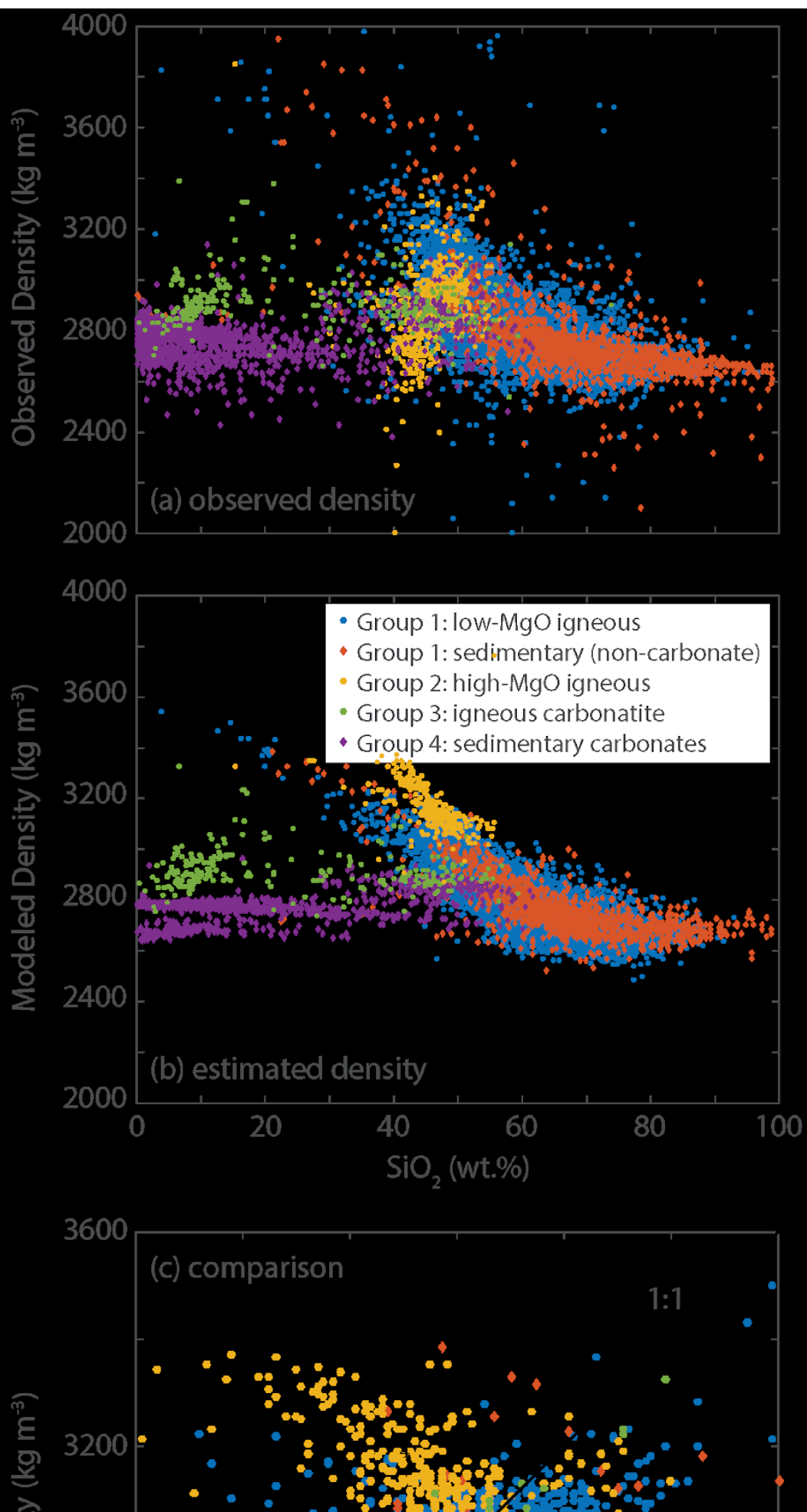
6.3 Isotropy vs anisotropy

- Isotropic properties (e.g., density) are direction independent.
- Anisotropic properties (e.g., thermal/electrical conductivity, elastic moduli) vary with direction.

Bulk density as an example. With volume fractions f_i and constituent densities ρ_i ,

$$\rho_{\text{bulk}} = \sum_i f_i \rho_i. \quad (2)$$

oindent*Transcription notes:* Ambiguities in the handwriting/OCR are marked with “??” to flag items needing your confirmation.



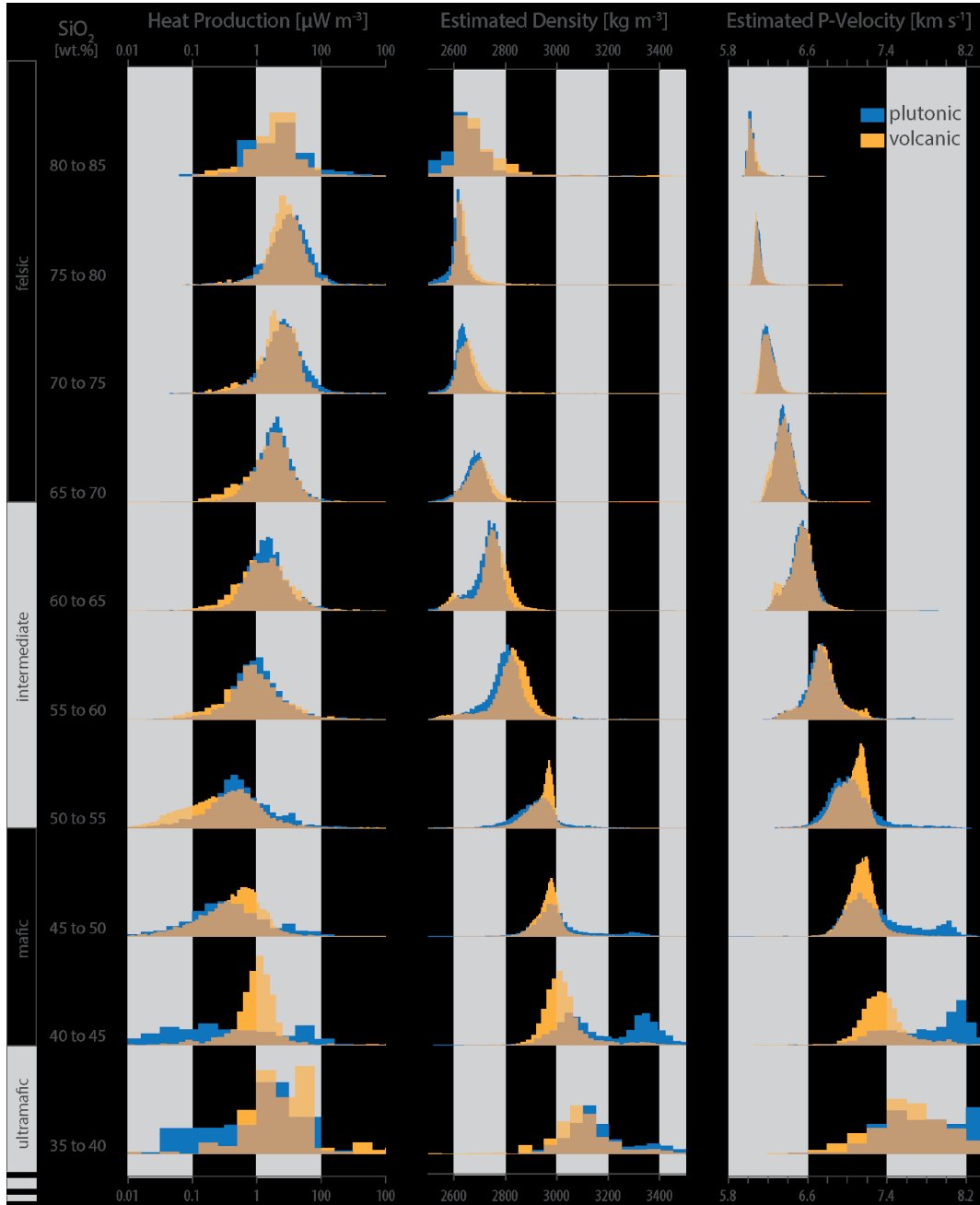


Figure 2: Compositional trends among properties (extracted from notes).

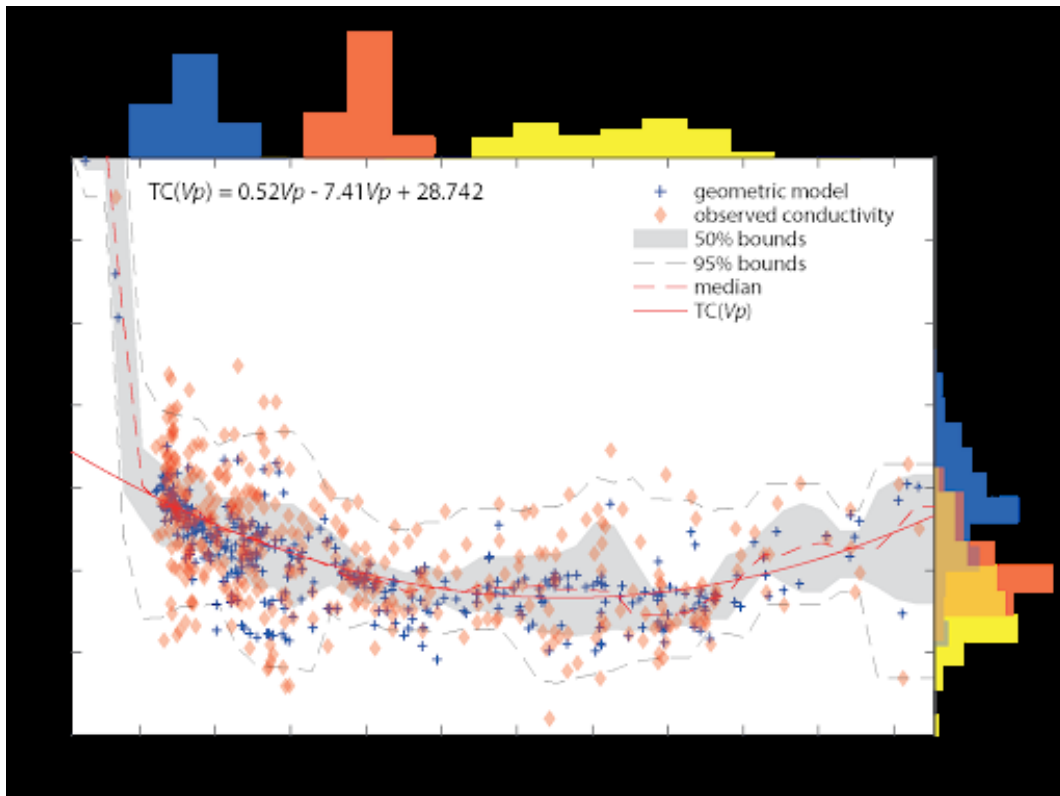


Figure 3: Thermal conductivity vs composition or v_p (extracted from notes).

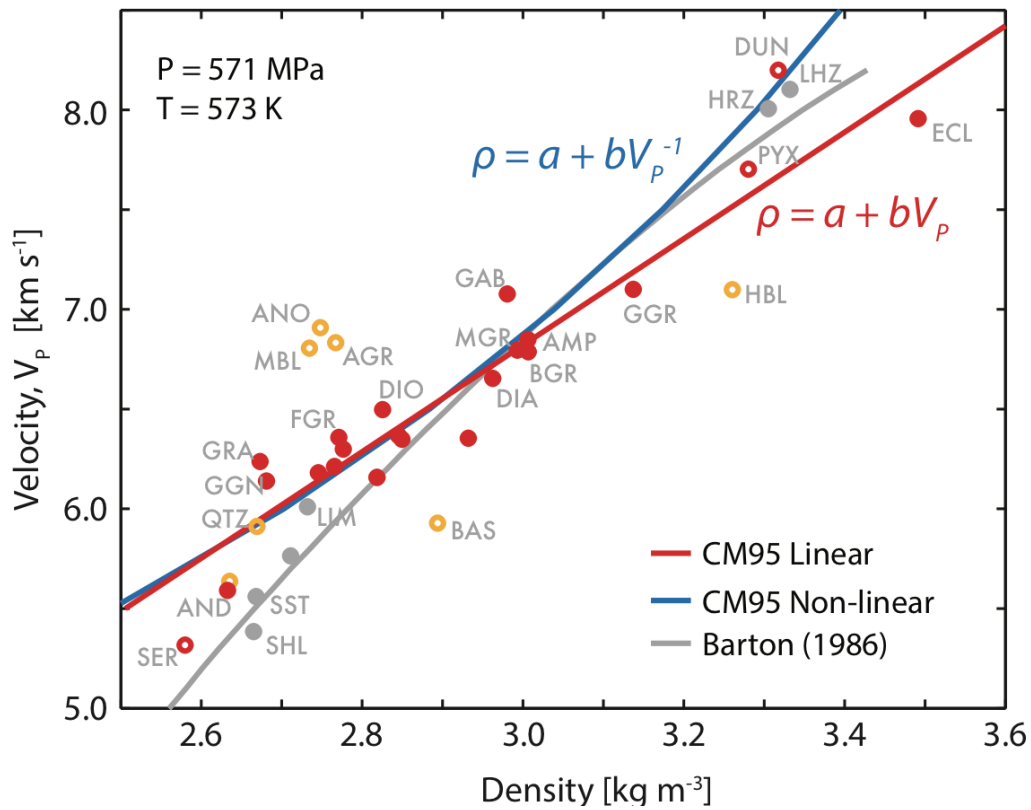


Figure 4: Seismic velocity vs density as composition proxy (extracted from notes).

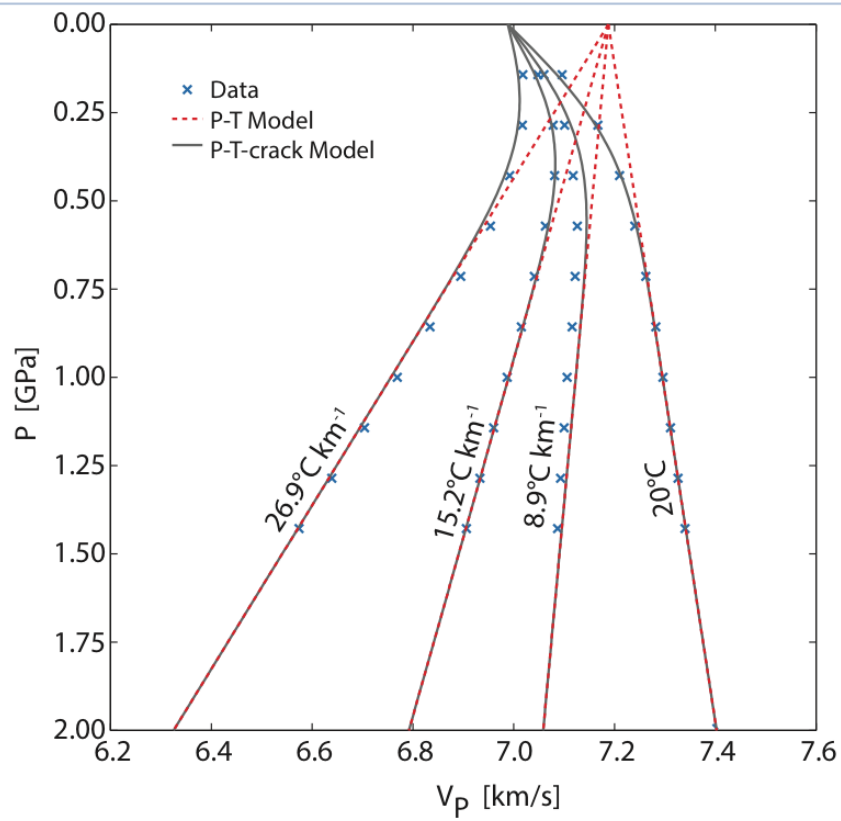


Figure 5: Velocity trends with P and T (extracted from notes).



Figure 6: Effect of porosity/fractures on velocity (extracted from notes).



Figure 7: Additional schematic/plot related to mixing or trends (extracted from notes).